

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



A i - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



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VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

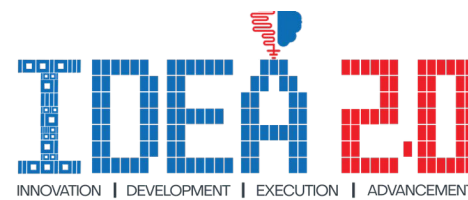


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DEPARTMENT OF
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PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME:		Team_Technexis	
PROBLEM STATEMENT TITLE:		PS3 — Tracking of Funds within Bank for Fraud Detection	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Arnav Shirwadkar	Male	Full Stack Web Dev
Member 2	Madhav Shukla	Male	CyberSecurity
Member 3	Sujalkumar Patil	Male	AIML

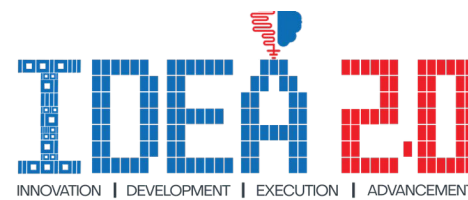


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IDEA TITLE

IntelliTrace: AI-Powered Fund Flow Intelligence for Banking Fraud Detection

Problem:

INR 13,930 Cr lost to banking fraud in FY2024; funds move through layered account chains invisible to siloed tools

1

Real-Time Graph Fund Tracking

Maps every rupee across
accounts, products,
branches and channels
as a live transaction
network

2

5 Suspicious Pattern Detectors

Smurfing, round-tripping,
rapid layering, dormant
activation & profile
mismatch — all encoded
as graph algorithms

3

Insider Fraud Signal Layer

Identifies the privileged
user who approved the
suspicious chain —
adding accountability to
traceability

4

Automated FIU Evidence Package

From detection to
regulatory submission in
under 15 minutes — fully
PMLA-compliant
STR/CTR output

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

01

Graph-First Architecture

Accounts as nodes, transactions as weighted edges.
Detects structural patterns invisible to row-by-row SQL queries — circular chains, hub accounts, velocity bursts.

04

Auto FIU Evidence Package

Structured STR/CTR documents formatted for FIU-IND under PMLA generated automatically.
Detection-to-filing in under 15 minutes vs. 2–4 analyst hours manually.

02

5 Encoded Detection Archetypes

Smurfing (clustering below INR 10L), Round-tripping (cycle detection), Rapid Layering (time-velocity), Dormant Activation, Profile Mismatch — each a distinct graph algorithm.

05

Cross-Domain Insider Fusion

Fund flow anomalies are correlated with the privileged user's behaviour score — if the same insider flagged in PSI approved the suspicious chain, risk is immediately elevated to CRITICAL.

03

Gen-AI Contextual Explanations

Claude API generates plain-English investigation summaries per flagged case — explaining WHY the pattern is suspicious and what action to take.

06

Live D3.js Force Graph

Investigators watch money move in real time on an interactive network graph. Click any node to drill into account history, risk score, and the complete transaction trail.

TECHNICAL APPROACH

DIAGRAM 1 — TECHNICAL ARCHITECTURE

System Components

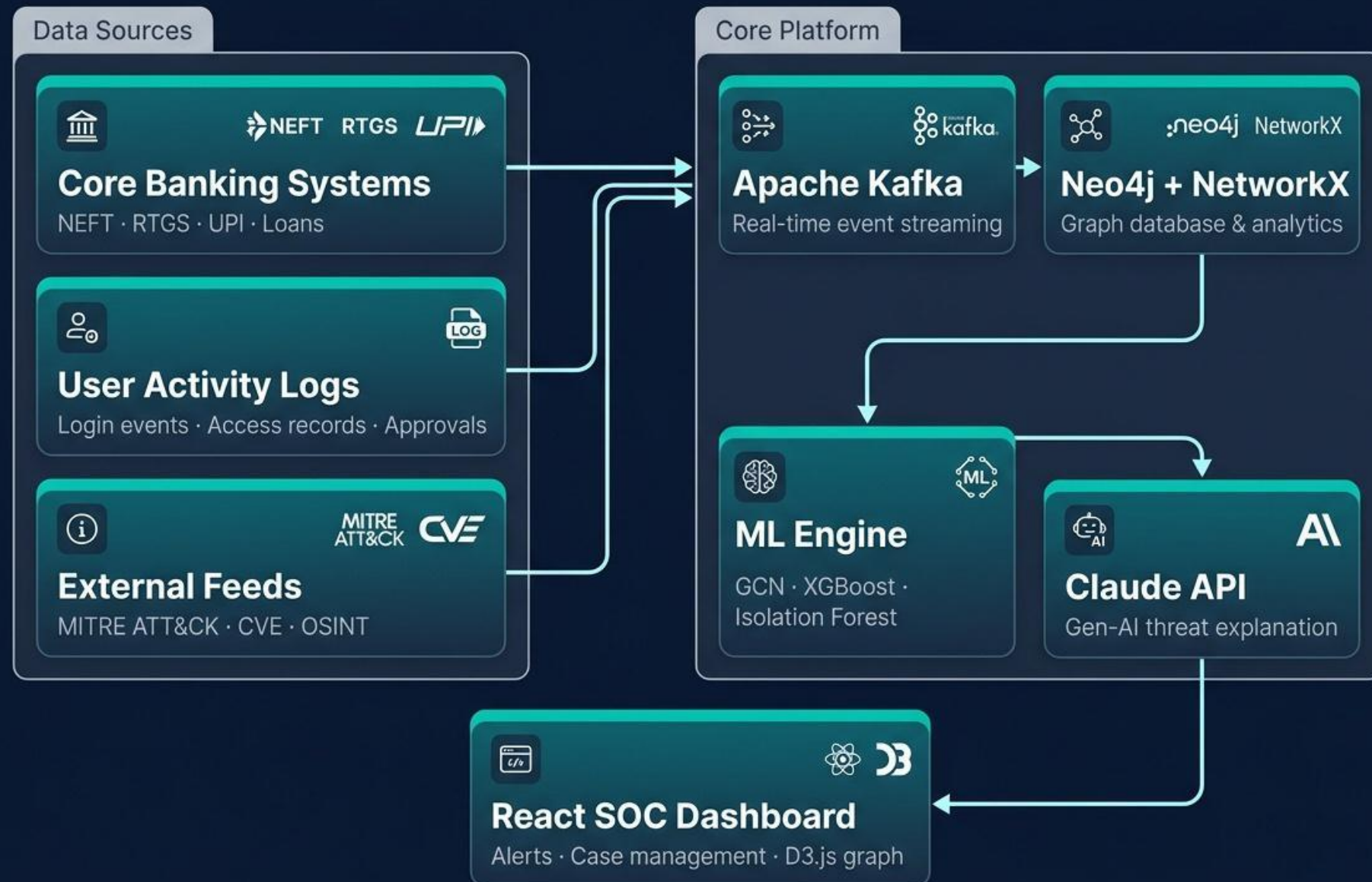


DIAGRAM 2 — DATA PIPELINE

5-STAGE FLOW



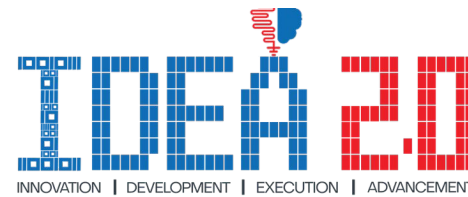


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FEASIBILITY & VIABILITY

Prepare Tools

FEASIBILITY

All technologies are production-proven in banking: Neo4j powers fraud detection at HSBC; Kafka runs at every major fintech

36-hour hackathon prototype is fully scoped on Docker Compose + PaySim synthetic data — no real banking data dependency

Modular design: start with read-only log ingestion (zero core banking API risk), add live integration in Phase 2

GCN + Isolation Forest are well-documented, reproducible models — not black boxes — satisfying regulatory explainability needs

MARKET & COMMERCIALISATION

! **Core banking API integration complexity**
→ Begin with read-only log ingestion; full API integration in Phase 2 after trust established

! **ML accuracy requires baseline data (30–60 days)**
→ Deploy in shadow mode initially; supervised feedback loop before autonomous alerting

! **DPDP Act compliance for behavioural monitoring**
→ Process only transaction metadata — never personal customer data; all models run on-premises

! **False positive risk damaging investigator trust**
→ Alerts require multi-signal corroboration (fund anomaly + second module) before escalation

IMPACT & BENEFITS



>> ANNUAL_LOSS_PREVENTION

INR 950 Cr+

Estimated annual savings across insider fraud, cyber breaches, and AML penalties.



>> INVESTIGATOR_EFFICIENCY

6X GAIN

Unified case management cuts a 4-hour, 3-analyst investigation to 40 minutes.



>> DETECTION_ACCURACY

90% REDUX

Alerts escalate only when a second module corroborates the same entity.



>> STR_FILING_TIME

15 MIN

From fraud detection to FIU-ready evidence package — down from 4 analyst hours.



Economic

- INR 90–240 Cr fraud loss prevention annually for a mid-sized bank
- INR 50–150 Cr regulatory penalty avoidance (AML fines)
- 20–35% lower cyber insurance premiums — saving INR 4–14 Cr/year



Social

- Disrupts money laundering pipelines that fund organised crime and terrorism
- Protects retail banking customers from fraud-driven account takeovers
- Builds public trust in the banking system through demonstrably proactive defence



Regulatory

- Meets PMLA 7-day STR filing obligation automatically — no analyst bottleneck
- Satisfies RBI Fraud Risk Management Directions 2024 in full
- Provides documented incident response evidence required by RBI Cyber Security Framework

BUSINESS MODEL

REVENUE MODEL

- Implementation Fee
 - **INR 8–12 Cr per bank**
 - Year 1 integration, training & customisation
- **SaaS Licence**
 - **INR 3–5 Cr / year**
 - Recurring per-bank annual subscription
- **FIU Reporting Module**
 - **Premium add-on**
 - Enhanced STR/CTR automation & regulator portal integration
- **Multi-Bank SaaS Platform**
 - **Year 2+ scale**
 - Single cloud instance serving cooperative & small finance banks

MARKET & COMMERCIALISATION

>> TARGET_MARKET

1,500+ Banking Entities

Private, Public, Neo-banks, and Cooperatives.

>> TOTAL_ADDRESSABLE_MARKET

\$2.1 Billion TAM

5-year projection across the BFSI security sector.

1

Lead with RBI Compliance Mandate

Banks facing RBI Fraud Risk Directions 2024 have pre-approved budget and compressed timelines — zero 'nice to have' objection

2

Pilot with One Large Private Bank

Read-only shadow mode in Month 1–3; first verified true positive case validates the platform

3

Partner with Core Banking Vendors

Infosys Finacle & Temenos pre-integrations unlock 80% of Indian banking market without bespoke API work

4

Expand to Cooperative & SFB Segment

Multi-tenant SaaS model makes IntelliTrace affordable for 1,500+ smaller banks — 32% CAGR AI investment tailwind



RESEARCH & REFERENCES

01

RBI Annual Report 2023–24

36,075 bank fraud cases · INR 13,930 Cr ·
300% YoY cyber fraud growth
<https://www.rbi.org.in/Scripts/AnnualReportPublications.aspx>

02

ACFE Report to the Nations 2024

5% of annual revenue lost to fraud ·
insider fraud stats and benchmarks
<https://www.acfe.com/report-to-the-nations>

03

Ponemon Institute Cyber Security Report 2024

3–7% market cap impact per breach event
· cyber insurance premium benchmarks
<https://www.ponemon.org>

04

MITRE ATT&CK Framework

Threat technique taxonomy used for
attack path modelling in Gen-AI module
<https://attack.mitre.org>

05

IBM Cost of a Data

Breach Report 2024

USD 9.5M average breach cost in
banking · 168-day avg detection window
<https://www.ibm.com/reports/data-breach>

06

PMLA 2002 (amended 2024) & FIU-IND Annual Report

STR/CTR obligations · 7-day filing window ·
AML penalty benchmarks
<https://fiuindia.gov.in>

07

PaySim Synthetic Financial Dataset

Realistic mobile money transaction simulation
data used for GCN model training
<https://www.kaggle.com/datasets/ealaxi/paysim1>

08

UN Office on Drugs & Crime — AML Statistics

2–5% of global GDP laundered annually = USD
800B–2T
<https://www.unodc.org/unodc/en/money-laundering>

Data sources, frameworks and
academic foundations